

How Transit and Green Tech Make Economic Geography More Local

The theme of winners and losers has been on my mind for the last few months, due to the politics of the Brooklyn bus redesign. In a rich country, practically every social or political decision is win-lose, even if the winners greatly outnumber the losers. It's possible to guarantee a soft landing to some of the losers, but sometime even the soft landing is disruptive, and it's crucial that backers of social change be honest with themselves and with the public about this. Overall, a shift from an auto-oriented society to a transit-oriented one and from dirty energy to clean energy is positive and must be pursued everywhere, but it does have downsides. In short, it changes economic geography in ways that make certain regions (like Detroit or the Gulf Cooperation Council states) redundant; it reorients economies toward more local consumption, so oil, gas, and heavy industry jobs would not be replaced with similar manufacturing or mining clusters but with slightly more work everywhere else in the world.

Dirty production is exportable

The United States has the dirtiest economy among the large developed countries, so it's convenient to look at average American behavior to see where the money that is spent on polluting products goes.

Nationally, [about 15.9% of consumer spending is on transportation](https://www.bls.gov/news.release/cesan.nr0.htm) (<https://www.bls.gov/news.release/cesan.nr0.htm>). The [vast majority of that is on cars](https://www.bls.gov/opub/ted/2017/consumer-spending-on-vehicles-averaged-8427-in-2016.htm?view_full) (https://www.bls.gov/opub/ted/2017/consumer-spending-on-vehicles-averaged-8427-in-2016.htm?view_full), 93.1% (that is, 14.7% of total consumer spending). The actual purchase of the car is 42% of transportation spending, or 6.7% of household spending. This goes to an industry that, while including local dealerships (for both new and used cars), mostly consists of auto plants, making cars in suburban Detroit or in low-wage Southern states and exporting them nationwide.

In addition to this 6% of consumer spending on cars, there's fuel. Around 3% of American household spending is on fuel for cars. Overall [US oil consumption in 2017 was 7.28 billion barrels](https://www.eia.gov/tools/faqs/faq.php?id=33&t=6) (<https://www.eia.gov/tools/faqs/faq.php?id=33&t=6>), which at [\\$52/barrel](https://www.statista.com/statistics/262858/change-in-opec-crude-oil-prices-since-1960/) (<https://www.statista.com/statistics/262858/change-in-opec-crude-oil-prices-since-1960/>) is 5% of household spending; the difference between 5 and 3 consists of oil consumed not by households. This is a total of about 2% of American GDP, which includes, in addition to household spending, capital goods and government purchases. This tranche of the American economy, too, is not local, but rather goes to the oil industry domestically (such as to Texas or Alaska) or internationally (such as to Alberta or Saudi Arabia).

Historically, when coal was more economically significant, it was exportable too. Money flowed from consumers, such as in New York and London, to producers in the Lackawanna Valley or Northeast England; today, it still flows to remaining mines, such as in Wyoming.

The same is true of much of the supply chain for carbon-intensive products. Heavy industry in general has very high carbon content for its economic value, which explains how the Soviet Union had high greenhouse gas emissions even with low car usage ([15.7 metric tons per capita in the late 1980s](http://cdiac.ess-dive.lbl.gov/ftp/trends/emissions/uss.dat) (<http://cdiac.ess-dive.lbl.gov/ftp/trends/emissions/uss.dat>)) – it had heavy industry just as the capital bloc did, but lagged in relatively low-carbon consumer goods and services. The economic geography of steel, cement, and other dirty products is again concentrated in industrial areas. In the US, Pittsburgh is famous for its historical steel production, and in general heavy manufacturing clusters in the Midwestern parts of the Rust Belt and in transplants in specific Southern sites.

All of these production zones support local economies. The top executives may well live elsewhere – for example, David Koch lives in New York and Charles Koch in Wichita (whose economy is based on airplane manufacturing and agriculture, neither of which the Kochs are involved in). But the working managers live in city regions dedicated to servicing the industry, the way office workers in the oil industry tend to live in Houston or Calgary,

and of course the line workers live near the plants and mines.

Clean alternatives are more local

The direct alternatives to oil, gas, and cars are renewable energy and public transportation. These, too, have some components that can be made centrally and exported, such as solar panel and rolling stock manufacturing. However, these components are a small fraction of total spending.

How small? Let's look at New York City Transit. [Its operating costs are about \\$9.1 billion a year](https://cms.fta.dot.gov/sites/fta.dot.gov/files/docs/ntd/66026/top-50-summary-and-complete-profile-set_1.pdf) (https://cms.fta.dot.gov/sites/fta.dot.gov/files/docs/ntd/66026/top-50-summary-and-complete-profile-set_1.pdf) as of 2016, counting both the subway and buses. Nearly all of this is wages, salaries, and benefits: \$7.3 billion, compared with only \$500 million for materials and supplies. This specifically excludes vehicle purchases, which in American transit accounting are lumped as capital costs. The total NYCT fleet is about 6,400 subway cars, which [cost around \\$2.3 million each](https://en.wikipedia.org/wiki/R179_(New_York_City_Subway_car)) ([https://en.wikipedia.org/wiki/R179_\(New_York_City_Subway_car\)](https://en.wikipedia.org/wiki/R179_(New_York_City_Subway_car))) and last 40+ years, and 5,700 buses, which [cost around \\$500,000 each](http://www.mta.info/press-release/nyc-transit/mta-purchase-120-new-buses) (<http://www.mta.info/press-release/nyc-transit/mta-purchase-120-new-buses>) and last 12 years, for a total depreciation charge of around \$600 million a year combined.

Compare this with cars: New York [has about 2 million registered cars](https://dmv.ny.gov/statistic/2017reinforce-web.pdf) (<https://dmv.ny.gov/statistic/2017reinforce-web.pdf>), but at the same average car ownership rate as the rest of the US, [845 per 1,000 people](https://www.statista.com/statistics/183505/number-of-vehicles-in-the-united-states-since-1990/) (<https://www.statista.com/statistics/183505/number-of-vehicles-in-the-united-states-since-1990/>), it would have 7.3 million cars. These 5.3 million extra cars [would cost \\$36,500 each today](http://www.businessinsider.fr/us/americans-spending-big-money-for-new-cars-2018-1) (<http://www.businessinsider.fr/us/americans-spending-big-money-for-new-cars-2018-1>), and last around 20 years, for a total annual depreciation charge of \$9.7 billion.

Put another way, total spending on vehicles at NYCT is one sixteenth what it would take to raise the city's car ownership rate to match the national average. Even lumping in materials and supplies that are not equipment, such as spare parts and fuel for buses, the total, \$1.1 billion, is one ninth as high as buying New Yorkers cars so that they can behave like Americans outside the city, and that's without counting the cost of fuel. In particular, there is no hope of maintaining auto plant employment by retraining auto workers to make trains, [as Michael Moore proposed in 2009](https://www.thetruthaboutcars.com/2009/06/michael-moore-on-gm-c11/) (<https://www.thetruthaboutcars.com/2009/06/michael-moore-on-gm-c11/>).

The vast majority of transit spending is then local: bus and train operations, maintenance, and local management. The same is true of capital spending, which goes to local workers, contractors, and consultants, and even when it is outsourced to international firms, the bulk of the value of the contract does not accrue to Dragados or Parsons Brinckerhoff.

Clean energy is similarly local. Solar panels can be manufactured centrally, but installing them on rooftops is done locally. Moreover, the elimination of carbon emissions coming from buildings has to come not just from cleaner electricity but also from reducing electricity consumption through passive solar construction. Retrofitting houses to be more energy-efficient is a labor-intensive task comprising local builders sealing gaps in the walls, windows, and ceilings.

Low-carbon economic production can be exported, but not necessarily from Detroit

A global shift away from greenhouse gas emissions does not mean just replacing cars and oil with transit and solar power. Transit is cheaper to operate than cars: in metro New York, [80.5% of personal transportation expenditure is still on cars](https://www.bls.gov/regions/new-york-new-jersey/news-release/consumerexpenditures_newyorkarea.htm) (https://www.bls.gov/regions/new-york-new-jersey/news-release/consumerexpenditures_newyorkarea.htm), and the rest is (as in the rest of the country) partly on air travel and not transit fare, whereas work trip mode shares in the metropolitan statistical area are 56% car, 31% transit. With its relatively high (for North America) transit usage, metro New York has the lowest share of household spending going to transportation, just 11.4%. This missing consumption goes elsewhere. Where does it go?

The answer is low-carbon industries. Consuming less oil, steel, and concrete means not just consuming more local labor for making buildings more efficient and running public transit, but also shifting consumption to less

carbon-intensive industries. This low-carbon consumption includes local purchases, for example going out to eat, or hiring a babysitter to look after the kids, neither of which involves any carbon emissions. But it also includes some goods that can be made centrally. What are they, and can they be made in the same areas that make cars and steel or drill for oil and gas?

The answer is no. First, in supply regions like the Athabaskan Basin, Dammam, and the North Slope of Alaska, there's no real infrastructure for any economic production other than oil production. The infrastructure (in the case of North America) and the institutions (in the case of the Persian Gulf) are not suited for any kind of manufacturing. Second, in real cities geared around a single industry, like Detroit or Houston, there are still lingering problems with workforce quality, business culture, infrastructure, and other necessities for economic diversification.

Take the tech industry as an example. The industry itself is very low-carbon, in the sense that software is practically zero-carbon and even hardware has low carbon content relative to its market value. Some individual tech products are dirty, such as Uber, but the industry overall is clean. A high carbon tax is likely to lead to a consumption shift toward tech. And tech as an industry has little to look for in Detroit and Houston. Austin has booming tech employment, but Houston does not, despite having an extensive engineering sector courtesy of the oil industry as well as NASA. The business culture in the space industry (which is wedded to military contracting) is alien to that of tech and vice versa; the way workers are interviewed, hired, and promoted is completely different. I doubt the engineers oil and auto industries are any more amenable to career change to software.

On the level of line workers rather than engineers, the situation is even worse. A manufacturing worker in heavy industry can retrain to work in light industry, or in a non-exportable industry like construction, but light industry has little need for the massive factories that churn out cars and steel. And non-manufacturing exports like tech don't employ armies of manufacturing workers.

In Germany the situation is better, in that Munich and Stuttgart may have little software, but they do have less dirty manufacturing in addition to their auto industries. It's likely that if global demand for cars shifts to a global demand for trains then Munich will likely keep thriving – it's the home of not just BMW and Man but also Siemens. However, the institutions and worker training that have turned southern Germany into an economically diverse powerhouse have not really replicated outside Germany. Ultimately, in a decarbonizing world, southern Germany will be the winner among many heavy industrial regions, most of which won't do so well.

There's no alternative to shrinkage in some cities

A shift away from fossil fuel and cars toward green energy and public transit does not have to be harsh. It can aim to give individual workers in those industries a relatively soft landing. However, two snags remain, and are unavoidable.

The first is that [some line workers have deliberately chosen poor working conditions in exchange for high wages](https://np.reddit.com/r/programming/comments/9it3ar/treating_the_bay_area_as_if_it_is_like_working_in/e6mvzqm/?context=3) (https://np.reddit.com/r/programming/comments/9it3ar/treating_the_bay_area_as_if_it_is_like_working_in/e6mvzqm/?context=3); the linked example is about oil rig workers in Alaska, but the same issue occurs in some unionized manufacturing and services, for example electricians get high wages but all suffer hearing loss by their 50s. It's possible to retrain workers and find them work that's at the same place on the average person's indifference curve between pay and work conditions, but since those workers evidently chose higher-pay, more dangerous jobs, their personal preference is likely to weight money more than work conditions and thus they're likely to be unhappy with any alternative.

The second and more important snag is the effect of retraining on entire regions. Areas that specialize to oil, gas, cars, and to some extent other heavy industry today are going to suffer economic decline, as the rest of the world shifts its consumption to either local goods (such as transit operations) or different economic sectors that have no reason to locate in these areas (such as software).

Nobody will be sad to see Saudi Arabia crash except people who are directly paid by its government. But the leaders of Texas and Michigan are not Mohammad bin Salman; nonetheless, it is necessary to proceed with

decarbonization. It's not really possible to guarantee the *communities* a soft landing. Governments all over the world have wasted vast amounts of money trying and failing to diversify from one sector (e.g. oil in the GCC states) or attract an industry in vogue (e.g. tech anywhere in the world). If engineering in Detroit and Houston can't diversify on its own, there's nothing the government can do to improve it, and thus these city regions are destined to become much smaller than they are today.

This is bound to have knock-on regional effects. Entire regions don't die quietly. Firms specializing in professional services to the relevant industries (such as Halliburton) will have to retool. Small business owners who've dedicated their lives to selling food or insurance or hardware to Houstonians and suburban Detroit white flighters will need to leave, just as their counterparts in now-dead mining towns or in Detroit proper did. Some will succeed elsewhere, just as many people in New Orleans who were displaced by Katrina found success in Houston. But not all will. And it's not possible to guarantee all of them a soft landing, because it's not possible to guarantee that every new small business will succeed.

All policy, even very good policy, has human costs. There are ways to reduce these costs, through worker retraining and expansion of alternative employment (such as retrofitting older houses to be more energy-efficient). But there is no way to eliminate these costs. Some people who are comfortable today will be made precarious by any serious decarbonization program; put another way, these people's entire livelihood depends on continuing to destroy the planet, and most of them are not executives at oil and gas companies. It does not mean that decarbonization should be abandoned or even that it should be pursued more hesitantly; but it does mean climate activists, including transit activists, have to be honest about how it affects people in and around polluting industries.

Written by [Alon Levy](https://pedestrianobservations.com/author/abstractnonsense/) (<https://pedestrianobservations.com/author/abstractnonsense/>) — Posted in [Cars](https://pedestrianobservations.com/category/transportation/cars/) (<https://pedestrianobservations.com/category/transportation/cars/>), [Development](https://pedestrianobservations.com/category/urbanism/development/) (<https://pedestrianobservations.com/category/urbanism/development/>), [Environmental Issues](https://pedestrianobservations.com/category/environmental-issues/) (<https://pedestrianobservations.com/category/environmental-issues/>), [Politics and Society](https://pedestrianobservations.com/category/politics-and-society/) (<https://pedestrianobservations.com/category/politics-and-society/>), [Regional Rail](https://pedestrianobservations.com/category/transportation/regional-rail/) (<https://pedestrianobservations.com/category/transportation/regional-rail/>), [Transportation](https://pedestrianobservations.com/category/transportation/) (<https://pedestrianobservations.com/category/transportation/>), [Urban Transit](https://pedestrianobservations.com/category/transportation/urban-transit/) (<https://pedestrianobservations.com/category/transportation/urban-transit/>), [Urbanism](https://pedestrianobservations.com/category/urbanism/) (<https://pedestrianobservations.com/category/urbanism/>)

31 comments



2018-10-23 - 11:26

[IAN Mitchell](https://www.linkedin.com/in/ian--mitchell/) (<https://www.linkedin.com/in/ian--mitchell/>)

“Cities geared around a single industry, like Detroit or Houston”
Hold up now, Alon.
Houston has a diversified economy.
3% of the city's economy is in oil and gas.

Diamonds are about as important to israel's economy as oil and gas is to Houston's.

Houston, having three times as many people as Austin, has more overall high-tech workers than Austin.

The largest employers in Houston are three world-class hospitals, regional grocer H-E-B, and Wal*Mart.

The big 3 are still the 3 largest employers in Metro Detroit, each of them employing nearly 100,000 people.

The largest oil company in Houston, ExxonMobil, has about 10,000 employees in the metro area.

Keep in Mind, Greater Houston has almost twice as many inhabitants and twice as many workers as Metro Detroit.

Houston's growth in the 20th century may have been fueled by oil (but also by affordability, safety, lack of toxic zoning laws, desirable climate), but Houston doesn't need oil to drive its continued growth to become America's third-largest city.

Austin's cool (I live here for a reason), but Houston isn't remotely similar to Detroit.



2018-10-23 - 11:38

[Alon Levy_ \(https://pedestrianobservations.wordpress.com/\)](https://pedestrianobservations.wordpress.com/)

Houston is really not diversified. As of the early 2010s, nine out of ten of its largest corporations were in the oil industry, broadly construed (e.g. Halliburton provides professional services to oil firms); the tenth was Continental, which is no longer there. It's like Koch Industries – before diversifying to the monstrosity it is today, it sold machines to oil firms that helped them drill more efficiently.

Wal-Mart retail is not an export product. It's a local service for other firms, which in Houston's case are in oil and gas and related industries.



2018-10-23 - 13:12

Diego Beghin

Wal-Mart retail is not an export product. It's a local service for other firms, which in Houston's case are in oil and gas and related industries.

Yes, when trying to evaluate economic diversity, we have to look at the “export” sector, what does Houston sell to other cities in the US or to other countries. If the export sector shuts down, there'll be no way to finance the local economy except through fiscal transfers.



2018-10-23 - 14:22

[IAN Mitchell \(https://www.linkedin.com/in/ian--mitchell/\)](https://www.linkedin.com/in/ian--mitchell/)

So you're going with “largest companies” by what metric? Clearly not employment. Market cap? Revenue?

Houston exports medical research. Houston exports professional services. United Airlines employs more people in Houston than any oil/gas sector company (and they didn't even make my list).

Metro Detroit is a region with 2.5 million workers, approximately 300,000 of whom work for three companies, in a sector that is an a broad-based long-term decline.

Houston is a region with 4.5 million workers, about 50,000 of whom work for the largest three oil and gas companies.

More people work for United Airlines than ExxonMobil in Houston. More people work for UT Medical center than for Shell in Houston.

More people work for HP, HCA, JPMorgan, or AT&T than work for Chevron, Conocophillips, or BP in Houston.

What does Houston export? Banking, professional services, medical research, higher education, transportation.

Not as much of those things as New York, Boston, and Chicago, but it's not a single-industry city by any means.

US Oncology, Sysco, Waste Management, Minute Maid, Houston has headquarters for huge companies that have absolutely nothing to do with Oil & Gas.

Furthermore, the largest investors in the green energy sector?
They're those exact same oil and gas companies.

<https://www.fool.com/investing/2018/06/04/big-oil-is-investing-billions-in-renewable-energy.aspx>
(<https://www.fool.com/investing/2018/06/04/big-oil-is-investing-billions-in-renewable-energy.aspx>).

Houston's not going anywhere.

Metro Detroit? It's going to stabilize at a lower population than it has today. Still has shrinking to do.
Chicago? Hasn't even reached the peak of its decline. Cook County has lost 100,000 per decade my entire lifetime, and the decline is hastening, not slowing.



2018-10-23 - 15:09

[Alon Levy \(https://pedestrianobservations.wordpress.com/\)](https://pedestrianobservations.wordpress.com/)

I'm basing this on a Fortune 500 list, so it's revenue, not employment. Oil and gas is generally a less concentrated sector than cars, which is how Houston has 9 out of its 10 biggest companies (as of, like, a decade ago) in that industry – there's All American Pipeline, Halliburton, and what not.

ExxonMobil is based in Irving and Shell is based in the Hague and London.

Chicago is in decline but not because of lack of economic diversification. It has a diverse economy, it's just weak in the growth industries and is the capital of a region in decline. There's a blog post I've wanted to write since before I started this blog – it came out of comments on the Urbanophile that are so old they're probably gone in Aaron's website migration – contrasting big and globalized cities. Chicago is big, but its fortunes are tethered to the Midwest's; Paris is big, but its fortunes are tethered to France's; Tokyo is vast, but its fortunes are tethered to Japan's. In contrast, New York, London, Frankfurt, and San Francisco are all global (well, Frankfurt and London are pan-European, but that's 700 million people serviced, which is close enough to global for government work).



2018-10-23 - 15:41

adirondacker12800

Chicago's MSA and CSA are growing.

https://en.wikipedia.org/wiki/Chicago_metropolitan_area#Demographics
(https://en.wikipedia.org/wiki/Chicago_metropolitan_area#Demographics).

People like to eat, Chicago will always be there.



2018-10-23 - 15:36
adirondacker12800

Metro Detroit is a region with 2.5 million workers, approximately 300,000 of whom work for three companies, in a sector that is an a broad-based long-term decline.

No it's not. Automation has been eliminating jobs slinging parts on assembly lines or making parts that arrive at an assembly line. People will still want cars. Lots of them. The parts are more or less the same for any kind of drive train. The people in transmission plants may be in for a lot of hurt but the rest of the car is more or less the same.



2018-10-23 - 15:58
[Alon Levy \(https://pedestrianobservations.wordpress.com/\)](https://pedestrianobservations.wordpress.com/)

People don't particularly want American cars, is the problem.



2018-10-23 - 16:23
adirondacker12800

North Americans don't particularly want cars. They want trucks and SUVs. Though many SUVs look suspiciously like minivans to me. The people in the plant making doohickeys for the whatchamacallits don't care if it going into a Ford or Toyota or Audi or BMW that is being assembled in North America. Or the people in the assembly plant. No one asks them who cuts their paycheck when they wanna spend the money.

<https://www.automobilemag.com/news/the-15-top-producing-american-car-plants-151801/>
(<https://www.automobilemag.com/news/the-15-top-producing-american-car-plants-151801/>)



2018-10-23 - 17:13
[Alon Levy \(https://pedestrianobservations.wordpress.com/\)](https://pedestrianobservations.wordpress.com/)

Yes, and you guys like pickups and SUVs because fuel costs \$2-3 per gallon whereas here it approaches \$2 per liter. The reason the American auto industry crashed so hard to begin with is that it got used to making giant-ass cars with chrome tailfins and then after the 1973 oil crisis it had no clue how to make small, fuel-efficient cars that non-paupers liked driving, whereas the Japanese car industry was used to this. In the late 1970s, GM asked for a few years of protectionism so that it could get its act together and compete with Toyota and Honda. 40 years later, nobody buys American over here unless it's the European Ford Focus, and Americans buy Hondas and Toyotas as much as they do Fords and Chevys. Protectionism is a hoot.



2018-10-23 - 18:26
adirondacker12800

The people in Applebee's don't care if you are paying the bill by making parts for a pickup truck or a sewing machine on wheels. Or just a plain old sewing machine. Though I don't think we've made sewing machines for a long time. Yes we do, I can still get parts for my 1920's overlock from the original manufacturer. Not that I've needed parts since 1970. But it's not a plain old sewing machine either.

They stopped putting chrome tailfins on them in the 60s. Last time I looked it wasn't 1973 anymore. It'd be really kewl to have another '59 Impala. Until it needs it's weekly maintenance. I still have the tools, except for the dwell meter/tachometer. It's handy knowing how to gap spark plugs, clean carburetors and fool around with the choke, the lawnmower and the snowblower still have quaint ignition systems. Back in July there was a severe thunderstorm that took out a mile of trees between here and the substation. The generator started on the third pull. I keep meaning to see if there is a propane conversion kit for it. Someday...

It's odd the way German carmakers have a large market share in Germany and French carmakers have a large market share in France and Japanese carmakers in Japan and Indian carmakers in the U.K. Silly North Americans buying automobiles that fit the market.

http://www.wsj.com/mdc/public/page/2_3022-autosales.html
(http://www.wsj.com/mdc/public/page/2_3022-autosales.html)



2018-10-24 - 04:01

Eric

In general, suburbs require cars rather than transit for getting around, due to their low density. These areas house a high fraction of the US population. This is a geometric issue, technological and economic changes will not affect it.

Cars may switch from gasoline to electric, but existing car companies' expertise at building everything except the battery and motor will likely keep them competitive in the electric car market.

The geometric issue which requires cars can only be avoided by mass population movement to denser housing (which will have to be built, they don't currently exist). And if people are moving on such a massive scale, why is it more disturbing if they move out of the Detroit area to other areas?

In short, I don't expect massive changes to Detroit due to decarbonization. Perhaps Detroit will be outcompeted by Asian car companies; that is an issue unrelated to decarbonization.



2018-10-24 - 14:26

adirondacker12800

Automobile companies have been making electric motors for over century – starter motors – and have pretty good grip on the concept. And generators and later alternators. They know what they are doing.



2018-10-23 - 14:08

adirondacker12800

Continental's headquarters evaporated. Houston is a United hub and those people are still there. They probably were the biggest chunk of Continental employees in Houston before the merger.



2018-10-23 - 14:42

[Alon Levy \(https://pedestrianobservations.wordpress.com/\)](https://pedestrianobservations.wordpress.com/)

The airline hub is not that big of a deal. Exhibit A: Detroit is Delta's #2 hub.



2018-10-23 - 15:43
adirondacker12800

Baggage handlers go out to eat, shop in Walmart and have a car and a mortgage. Well, they are union, they probably shop in Target or Kohl's.



2018-10-23 - 12:11
Reedman Bassoon

"In Germany the situation is better" ???

Germany is shutting-down its carbon-free nuclear power plants and building more lignite coal power plants (it has re-settled over 20k people to expand coal mining).

<https://www.wired.com/story/if-germany-cant-quit-coal-can-anyone-else/> (<https://www.wired.com/story/if-germany-cant-quit-coal-can-anyone-else/>)

<https://qz.com/1389135/germany-is-razing-a-12000-year-old-forest-to-expand-a-coal-mine/> (<https://qz.com/1389135/germany-is-razing-a-12000-year-old-forest-to-expand-a-coal-mine/>)

<http://www.spiegel.de/international/business/energy-transition-blocked-by-brown-coal-a-1179537.html> (<http://www.spiegel.de/international/business/energy-transition-blocked-by-brown-coal-a-1179537.html>)



2018-10-23 - 14:23
[IAN Mitchell \(https://www.linkedin.com/in/ian--mitchell/\)](https://www.linkedin.com/in/ian--mitchell/)

France is similarly making a move to dirtier energy.



2018-10-23 - 14:41
[Alon Levy \(https://pedestrianobservations.wordpress.com/\)](https://pedestrianobservations.wordpress.com/)

Is it? Macron has said he's for nuclear power and is not decommissioning plants.



2018-10-28 - 10:43
Max Wyss

However, Germany had some days earlier this year, where the whole energy consumption allegedly was covered with "renewable" sources.



2018-10-23 - 13:27
Diego Beghin

This is why pushing for a strong social safety net together with an anti-climate change policy makes a lot of sense. When the economy has to shift gears very fast, it's good to make a job loss have less dramatic consequences. It's kinder to the workers, who made their career choices with a very different set of economic incentives, and it reduces their opposition to green policies.



2018-10-23 - 15:11
[Alon Levy \(https://pedestrianobservations.wordpress.com/\)](https://pedestrianobservations.wordpress.com/)

That works for the working class, but there's an entire petite bourgeoisie that's never getting an adequate social safety net (what are you going to do, public works for shopkeepers?).



2018-10-23 - 16:20

Diego Beghin

A UBI. Granted, it sucks, but it's better than nothing. The goal is to soften the opposition, not to eliminate it, that would be an unrealistic goal.



2018-10-23 - 17:47

[Desmond BLIEK \(@desmondbliek\)](https://twitter.com/desmondbliek) (<http://twitter.com/desmondbliek>)

Alon: Houston has an interesting dynamic and certainly has scale, but where is Calgary at and what should that city be doing/contemplating?



2018-10-24 - 02:09

[Alon Levy](https://pedestrianobservations.wordpress.com/) (<https://pedestrianobservations.wordpress.com/>)

Calgary is 100% oil-oriented, much more so than Houston. So decarbonization is likely to make it much less economically attractive than it is today. There's a reason it's the home of Canada's climate denialist politics (Stephen Harper, sigh).



2018-10-24 - 23:24

Josh

I am not convinced that transportation spending is lower in transit areas than car areas. Yes, personal spending is less, but transit is more government subsidized than cars (car gas tax spending is included in personal transportation expenditures, but metro NY MTA payroll tax is not). Capital spending is entirely government financed and even operating expenses aren't covered by fares. Definitely agree that is more localized, on labor, as opposed to imported cars, but not that it is less, such that there is room to import more. Also not included is increased commute times as a form of implied labor supplied by people from all industries.



2018-10-25 - 03:53

[Alon Levy](https://pedestrianobservations.wordpress.com/) (<https://pedestrianobservations.wordpress.com/>)

Of course there are subsidies (but for roads, too) – but I do take total operating costs into account (and also a portion of capital, but even on the widest definition it's on the order of one half the operating costs). The question is mainly what denominator you pick. If the denominator is the difference between New York's number of cars and how many cars it would have at rest-of-country travel behavior, then you get around \$1,700 in operating costs per missing car and on the order of \$900 in capital costs. This is in the context of unusually high costs, though – at the cost per car-km of Chicago, Paris, London, San Francisco, or Tokyo, the subway operating costs in New York would be barely half what they are today.

The longer commutes are real, but they're partly an issue of city size (Los Angeles has pretty long commutes too, if not so long as New York's), partly hedonic adjustment for being able to sit on a train and read, and partly compensation for having shorter non-work trips because you can walk to a lot of retail.



2018-10-27 - 13:38

[electricangel \(http://gravatar.com/electricangel1978\)](http://gravatar.com/electricangel1978)

Thanks for this analysis, Alon. You've done this a few times. Take readily available numbers and calculate them to show new insights.

Clearly, even at the cost of East Side Access the transit system confers a multi-billion dollar benefit on NYC. It would be nice if the little people could collect a larger share of the decreased transit depreciation bonus, of course.

2019-08-23 - 19:59

Mb

Hard to believe these regions dont tend to go along with a political philosophy of die... fuck this author.

2019-09-12 - 22:27

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2021-02-17 - 20:22

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